Pierrick BOUGAULT

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Personal Website

PROFILE

Industrial engineer and researcher building and evaluating multi-agent LLM systems for manufacturing quality, applied to aircraft production. Combines hands-on Python, RAG and agentic development with lean experience at Airbus, Renault and FANUC.

EXPERIENCE & APPLIED AI PROJECTS

TSINGHUA UNIVERSITY *Since 2024*
PhD Researcher – Applied Human-LLM Systems Beijing, CN

- Built a bilingual multi-agent LLM assistant for aerospace quality meetings (QRQC): speech-to-text, hybrid-RAG chatbot on a Neo4j knowledge graph, and agents updating records under human validation.
- Proof-of-value with 78 participants vs. paper process: effectiveness 84.9% $\rightarrow$ 97.2%, meeting time -38%, lower workload, usability rated acceptable.
- Optimized cost by routing each task to a suitable model tier: about USD 0.12 per meeting.

AIRBUS CHINA *August 2024*
Invited AI Workshop Lead Tianjin, CN

- Invited by the CIO to design and solely run an AI workshop for 40 engineers.
- Presented AI concepts and aerospace use cases, then assessed participants' own use cases and feasibility.

AIRBUS *September to November 2023*
Engineering Intern at AOS (Airbus Operating System) Beijing, CN

- Support for the implementation of lean tools and methodologies.
- Analysis of industrial performance data.

RENAULT ELECTRICITY *April to July 2022*
Process Reliability Intern on the Assembly Line Douai, FR

- Cut AGV-related line downtime by 85%, delivering €400k in annual savings for the plant.
- Redefined the workshop's autonomous maintenance standards.

FANUC *May to July 2021*
Industrial Robotics Intern Marnaz, FR

- Developed a robotic demonstration cell promoting the RMI (Remote Motion Interface) option.

EDUCATION & HONORS

Tsinghua University, CN, Ph.D. Engineering and Management Science – *Xinghua Excellence Scholarship* 2024 – 2028
Tsinghua University, CN, M.Sc. Industrial Engineering – *China Scholarship Council (CSC) Scholarship* 2022 – 2024
IMT Nord Europe (formerly Mines de Douai), FR, M.Eng. General Engineering 2018 – 2024
INSEAD, FR, Doctoral Summer Camp, Strategic Management as the Design of Complex Adaptive Systems 2026
Summer Schools: IMF & Renmin University, International Finance (2025) – INSEAD, Innovation and Leadership (2023)

SELECTED PUBLICATIONS

- **Don't Lose the Human Touch: Effects of an LLM-Based Assistant on Team Performance and Workload in Lean Manufacturing Meetings**. First author, submitted to *Applied Ergonomics*, 2026.
- **Designing Human-LLM Systems: A Review of Current State, Limitations, and Guidelines to Better Interaction**. First author, HCII 2026, Springer LNCS.
- **Industrial Applications of AI in Aircraft Manufacturing: A PRISMA Systematic Literature Review**. First author, Prod. Eng. Archives, 2026.

SKILLS

GenAI & LLM: Multi-agent systems, RAG, LangChain, Neo4j knowledge graphs, Prompt engineering, LLM evaluation
Programming: Python (primary), R, Matlab & Simulink, C, Java
Business & Industry: AI use-case discovery, Executive workshops, ROI quantification, Lean, Problem solving
Languages: Native in French, Fluent in English and Spanish, Intermediate in Chinese
Activities: VP, Franco-Chinese Association (Tsinghua) – CrossFit, Soccer, Judo (14 years) – Guitar, Bass, Drums